

The polluted atmosphere as a shallow domain



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The shallow domain concept

- Natural, large-scale geophysical fluids share the shallow domain characteristics
- "almost" two-dimensional set $\rightarrow$ we can significantly simplify the model in the vertical direction
- a domain that is excessively greater horizontally than in the vertical dimension $\rightarrow$ leading to the classic hydrostatic approximation of the Navier-Stokes equations

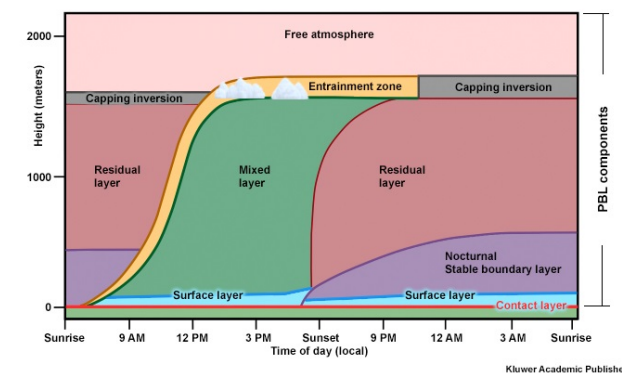


Fig. 1: A local slice of the atmosphere. Picture credit: mountwashington.org

The main ideas and tools in this scenario:

- introduce the aspect ratio $\epsilon = \frac{\text{characteristic depth}}{\text{characteristic width}}$
- anisotropic viscosity hypothesis,
- rescaling, $x = x_1, y = x_2, z = \epsilon x_3$.

Correspondingly, we rescale

- the kinematic terms: $v_3 = \epsilon u_3^\epsilon$,
- the viscosity terms: $\nu_z = \epsilon^2 \nu_3$,
- the diffusivity terms: $K_x = \epsilon K_1, K_y = K_2, K_z = \epsilon^2 K_3$.

The limit model

In the vertical dimension we arrive to the hydrostatic approximation.

$$\partial_t u_1 + \mathbf{u} \cdot \nabla u_1 - \Delta_v u_1 - \alpha u_2 + \partial_1 p = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t u_2 + \mathbf{u} \cdot \nabla u_2 - \Delta_v u_2 - \alpha u_1 + \partial_2 p = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_3 p = 0 \text{ in } \Omega \times (0, T)$$

$$\nabla \cdot \mathbf{u} = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t C + \mathbf{u} \cdot \nabla C = K_2 \frac{\partial^2 C}{\partial x_2^2} + K_3 \frac{\partial^2 C}{\partial x_3^2} + S \text{ in } \Omega \times (0, T)$$

$$u_1 = u_2 = u_3 n_3 = 0 \text{ on } \Gamma \times (0, T)$$

$$u_i(\cdot, t = 0) = u_{0i}, C(\cdot, t = 0) = C_0 \text{ in } \Omega, i = 1, 2$$

$$C = 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0, T), \partial_2 C = \partial_3 C = 0 \text{ on } \Gamma_G \times (0, T), \partial_1 C \in L^2 L^2(0, T; \Gamma_G)$$

A priori estimates, weak convergence

The weak form of the anisotropic system has the following structure.

$$\begin{aligned} & \int_0^T -(\mathbf{u}_H^\epsilon, \partial_t \mathbf{v}_H) + (\nabla_v \mathbf{u}_H^\epsilon, \nabla_v \mathbf{v}_H) - (\mathbf{u}_H^\epsilon, (\mathbf{u}^\epsilon \cdot \nabla) \mathbf{v}_H) + (b(\mathbf{u}_H^\epsilon), \mathbf{v}_H) dt \\ & + \epsilon \beta \left[\int_0^T (u_3^\epsilon, v_1) - (u_1^\epsilon, v_3) \right] dt \\ & + \epsilon^2 \left[\int_0^T -(u_3^\epsilon, \partial_t v_3) + (\mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon, v_3) + (\nabla_v u_3^\epsilon, \nabla_v v_3) dt \right] \\ & + \int_0^T -(C^\epsilon, \partial_t \tilde{C}) - (\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C}) + \\ & \epsilon K_1 (\partial_1 C^\epsilon, \partial_1 \tilde{C}) - \epsilon K_1 [(\partial_1 C^\epsilon) \tilde{C}]_{\Gamma_G} + K_2 (\partial_2 C^\epsilon, \partial_2 \tilde{C}) + K_3 (\partial_3 C^\epsilon, \partial_3 \tilde{C}) dt = \\ & = -(\mathbf{u}_{0H}^\epsilon, \mathbf{v}_H(0)) - \epsilon^2 (u_{03}^\epsilon, v_3(0)) - (C^\epsilon(0), \tilde{C}(0)) + \int_0^T (S^\epsilon, \tilde{C}) dt \end{aligned}$$

The main steps for proving the weak convergence are the following.

- Energy inequality.
- A priori estimates: $L_t^\infty L_x^2$ boundedness for C^ϵ ; $L_t^2 L_x^2$ boundedness for $\sqrt{\epsilon} \partial_1 C^\epsilon, \partial_2 C^\epsilon, \partial_3 C^\epsilon$.
- Weak convergence of the linear terms.
- Weak convergence of $(\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C})$

The anisotropic equations

$$\partial_t u_1^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_1^\epsilon - \Delta_v u_1^\epsilon - \alpha u_2^\epsilon + \epsilon \beta u_3^\epsilon + \partial_1 p^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t u_2^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_2^\epsilon - \Delta_v u_2^\epsilon + \alpha u_1^\epsilon + \partial_2 p^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\epsilon^2 \{ \partial_t u_3^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon - \Delta_v u_3^\epsilon \} - \epsilon \beta u_1^\epsilon + \partial_3 p^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\nabla \cdot \mathbf{u}^\epsilon = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t C^\epsilon + \mathbf{u}^\epsilon \cdot \nabla C^\epsilon = \epsilon K_1 \frac{\partial^2 C^\epsilon}{\partial x_1^2} + K_2 \frac{\partial^2 C^\epsilon}{\partial x_2^2} + K_3 \frac{\partial^2 C^\epsilon}{\partial x_3^2} + S_\epsilon \text{ in } \Omega \times (0, T)$$

$$\mathbf{u}^\epsilon = 0 \text{ on } \Gamma \times (0, T)$$

$$\mathbf{u}^\epsilon(\cdot, t = 0) = \mathbf{u}_0, C^\epsilon(\cdot, t = 0) = C_0 \text{ in } \Omega$$

$$C^\epsilon = 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0, T), \partial_2 C^\epsilon = \partial_3 C^\epsilon = 0 \text{ on } \Gamma_G \times (0, T), \partial_1 C^\epsilon \in L^2 L^2(0, T; \Gamma_G)$$

The original equations We use the incompressible Navier-Stokes equations in all three space dimensions for the fluid velocity.

The main theorem

- We provide an expansion of the convergence result of [Azérad and Guillén, 2001].
- The main theorem of their article is for the case of the Navier-Stokes equations of the ocean, without the inclusion of salinity.
- We prove that the hydrostatic approximation is still justifiable with the additional pollution concentration effect.
- The theorem is valid locally (see graph below).

Theorem

Let $\mathbf{u}_0 \in L^2(\Omega)^3$, with $\nabla \cdot \mathbf{u}_0 = 0, \mathbf{u}_0 \cdot \mathbf{n} = 0$ on $\partial\Omega$; and let $C(0) \in L^2(\Omega)^3$, $C(0) = 0$ on $\partial\Omega$; there exists a weak solution $(\mathbf{u}, C)$ of the hydrostatic equations of the polluted atmosphere, obtained as a limit of weak solutions $(\mathbf{u}^\epsilon, C^\epsilon)$ of the anisotropic equations as the aspect ratio ϵ tends to 0.

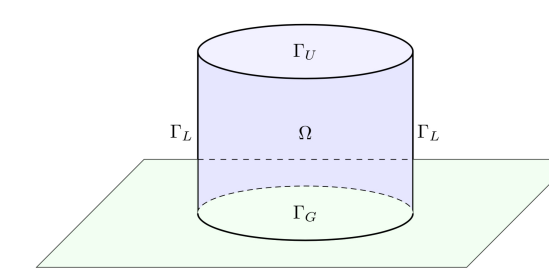


Fig. 2: A local slice of the atmosphere

We need a compactness result to obtain $u_i^\epsilon C^\epsilon \rightarrow u_i C$. For this we show that

- we have a bound for the time derivative of the concentration:

$$\frac{\partial C^\epsilon}{\partial t} = -\mathbf{u}^\epsilon \nabla C^\epsilon + \sqrt{\epsilon} K_1 \partial_1 \partial_1 \sqrt{\epsilon} C^\epsilon + K_2 \partial_2 \partial_2 C^\epsilon + K_3 \partial_3 \partial_3 C^\epsilon - S^\epsilon,$$

Eventually we have $\frac{\partial C}{\partial t} \in L^1 W^{-1,1}$,

- we have an equicontinuity for velocity: $\|u_i^\eta - u_i^\eta(\cdot + \xi, t)\|_{L^2 L^2} \rightarrow 0$ as $\xi \rightarrow 0$.

Combining these results allow us to apply a compactness theorem that provides the convergence.

Additional results

- Apart from the convergence theorem we also provide an existence theorem for the extended system.
- We have generalized the main theorem for the seaside case, when there is interaction between water and air $\rightarrow$ different BCs.

References

[Azérad and Guillén, 2001]Azérad, P. and Guillén, F. (2001). Mathematical justification of the hydrostatic approximation in the primitive equations of geophysical fluid dynamics. *SIAM Journal on Mathematical Analysis*, 33(4):847–859.

